**MATH 3603–Homework 2**

Due: 8-Oct-2026

**1.** Let $X$ represent the difference between the number of heads and the number of tails obtained when a coin is tossed $n$ times. (i):What are the possible values of $X$? (ii): If the coin is assumed fair, then, for $n = 3$, write up the probability mass function of $X$.

**2.** Let the probability density of $X$ be given by $f(x) = c(4x - 2x^2)$ for $0 < x < 2$, and 0 otherwise.

(a) What is the value of $c$?

(b) Compute $\mathbb{P}(1/2 < X < 3/2)$

(c) Compute $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

**3.** Let $X_1, X_2, ..., X_n$ be independent random variables with the same distribution function $F(x)$. What is the distribution function of $\max\{X_1, \ldots, X_n\}$ and $\min\{X_1, \ldots, X_n\}$?

**4.** Suppose that one year has 365 days. There are 100 people in a group, whose birthdays are independent, and each birthday is equally likely to be any day of a year. Let $X$ be the total number of the distinct birthdays in this group. Compute $\mathbb{E}X$ and $\mathrm{Var}(X)$. (Hint: write $X$ as a sum of random variables.)

**5.** A point $(X, Y)$ is uniformly distributed within the disk of radius 1. That is, its density is

$$f(x,y) = \begin{cases} C, & 0 \le x^2 + y^2 \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) What is the value of $C$? (ii) Compute the marginal probability density of $X$.

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P1 (i)

$$X \in \{-n, -n+2, \cdots, n\}$$

(ii) Let $H$ be the number of heads.

$\therefore H \sim Bin(3, \frac{1}{2})$.

$X = 2H - 3$.

$$P_X(x) = P_H\left(\frac{x+3}{2}\right)$$

$$= \binom{3}{\frac{x+3}{2}} \cdot \left(\frac{1}{2}\right)^3 \quad \text{when } x \in \{-3, -1, 1, 3\}$$

$$\therefore P_X(x) = \begin{cases} \binom{3}{\frac{x+3}{2}} \cdot \left(\frac{1}{2}\right)^3, & \text{when } x \in \{-3, -1, 1, 3\} \\ 0 & \text{otherwise.} \end{cases}$$

P2 (a) Note that $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\therefore \int_0^2 c \cdot (4x - 2x^2)\,dx = 1$$

$$c = \frac{1}{\int_0^2 (4x - 2x^2)\,dx} = \frac{3}{8}$$

(b) $P(\frac{1}{2} < X < \frac{3}{2}) = \int_{\frac{1}{2}}^{\frac{3}{2}} f(x)\,dx$

$$= \frac{3}{8}\int_{\frac{1}{2}}^{\frac{3}{2}} (4x - 2x^2)\,dx$$

$$= \frac{11}{16}$$

(c) $E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\,dx$

$= \int_0^2 x \cdot \frac{3}{8}(4x - 2x^2)\,dx$

$= \frac{3}{4}\int_0^2 (2x^2 - x^3)\,dx$

$= 1$

$E(X^2) = \int_{-\infty}^{\infty} (x^2) \cdot f(x)\,dx$

$= \frac{3}{4}\int_0^2 (2x^3 - x^4)\,dx$

$= \frac{6}{5}$

$\therefore Var(X) = E(X^2) - [E(X)]^2 = \frac{6}{5} - 1 = \frac{1}{5}$.

P3. Let $A = \max_i X_i$

$B = \min_i X_i$

$F_A(x) = P(A \le x)$

$= P(\max_i X_i \le x)$

$= P(X_1 \le x, X_2 \le x, \dots)$

$= F(x)^n$

$F_B(x) = P(B \le x)$

$= P(\min_i X_i \le x)$

$= 1 - P(\overline{\min_i X_i \le x})$

$= 1 - P(B \ge x)$

$= 1 - (1 - F(x))^n$

P4 Let $X_i = \begin{cases} 1 & , \text{if anyone's birthday is } i\text{-th day} \\ 0 & , \text{otherwise.} \end{cases}$

$\therefore X = \sum_{i=1}^{365} X_i$.

$E(X_i) = 1 - \left(\frac{364}{365}\right)^{100} = p$

$\therefore E(X) = \sum_{i=1}^{365} E(X_i) = 365\left(1 - \left(\frac{364}{365}\right)^{100}\right) = 365p$

$E(X_i X_j) = P(X_i = X_j = 1)$

$= 1 - P(X_i = 0) - P(X_j = 0) + P(X_i = X_j = 0)$

$= 1 - 2\left(\frac{364}{365}\right)^{100} + \left(\frac{363}{365}\right)^{100} = q$ when $i \neq j$

$E(X_i^2) = E(X_i)$

$\therefore E(X^2) = E\left[(\Sigma X_i)^2\right]$

$= \Sigma E(X_i^2) + 2 \sum_{1 \le i < j \le 365} E(X_i X_j)$

$= 365p + 365 \cdot 364\, q$

$\therefore \mathrm{Var}(X) = E(X^2) - (E(X))^2$

$= 365p + 365 \cdot 364 q - 365^2 p^2$

$= 365\left[-365 \cdot \left(\frac{364}{365}\right)^{200} + 2 \cdot \left(\frac{363}{365}\right)^{100} + 364 \cdot \left(\frac{363}{365}\right)^{100}\right]$

Q5. (i) $1 = \int_{R^2} f(x,y)\,dxdy$

$= \int_{R^2} C \cdot dxdy$

$= \int_0^{2\pi}\int_0^1 C R \cdot dR d\theta$, where $\begin{cases} x = R\cos\theta \\ y = R\sin\theta \end{cases}$

$= \int_0^{2\pi} \frac{C}{2} d\theta = \pi C$

$\therefore C = \frac{1}{\pi}$

(ii)

when $x \in [-1, 1]$

$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

$= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} f(x,y)\,dy$

$= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\,dy$

$= \frac{2\sqrt{1-x^2}}{\pi}$

when $x^2 > 1$, then $x^2 + y^2 \geq x^2 > 1$

i.e. $f(x,y) = 0$

$\therefore f_X(x) = \int_{-\infty}^{\infty} 0\,dy = 0$.

$\therefore f_X(x) = \begin{cases} \frac{2\sqrt{1-x^2}}{\pi}, & x \in [-1,1] \\ 0, & \text{otherwise.} \end{cases}$